

Weather Pattern Analysis

Introduction:

This report shows an exploratory data analysis of a historical weather dataset obtained from 'Kaggle'. Here the dataset contains 96,453 hourly weather recording that has been captured between 2006 and 2016 which covers 12 variables including,

- Temperature
- Humidity
- Wind speed
- Visibility
- Atmospheric pressure.

So the objective of this analysis is to understand how weather variables behave by identifying their patterns and outliers and explore the relationship between different conditions.

SECTION 1 : Understanding the data

The dataset consists of both numerical and categorical variables.

Numerical variables	Categorical variables
Temperature(C)	Weather Summary
Apparent Temperature (C)	Precipitation Type
Humidity	
Wind Speed(km/h)	
Visibility(km)	
Pressure(millibars)	

Hence before performing any analysis the data should be loaded and inspected to understand its structure and confirm that there were no major issues with the format.

SECTION 2: Statistical summary

The following table represents the key stats that is been calculated for each numerical variable. This includes mean, median, mode , std, min and max, 3 quartile values (Q1, Q2, Q3).

Variable	Mean	Median	Mode	Std	Min	Max	Q1	Q3
Temperature(C)	11.93	12.00	7.22	9.55	-21.82	39.91	4.69	18.84
Apparent Temperature(C)	10.86	12.00	12.78	10.70	-27.72	39.34	2.31	18.84
Humidity	0.73	078	0.93	0.20	0.00	1.00	0.60	0.89
Wind Speed(km/h)	10.81	9.97	3.22	6.91	0.00	63.85	5.83	14.14
Visibility(km)	10.35	10.05	9.98	4.19	0.00	16.10	8.34	14.81
Pressure(mb)	1003.24	1016.45	0.00	116.97	0.00	1046.38	1011.90	1021.09

Key observation from the above stats:

- Temperature ranges from -21.82°C to 29.91°C ; avg 12°C - four season climate typically.
- Humidity avg 73% but most commonly recorded at 93%.
- Wind avg 10.81 km/h but peaks at 63.85 km/h.
- Pressure has the highest variation from zero reading which indicates the possible sensor errors may be.
- Visibility drops to 0 km some days as it indicates fog cum heavy storm conditions.

SECTION 3 : Skewness and Outlier analysis

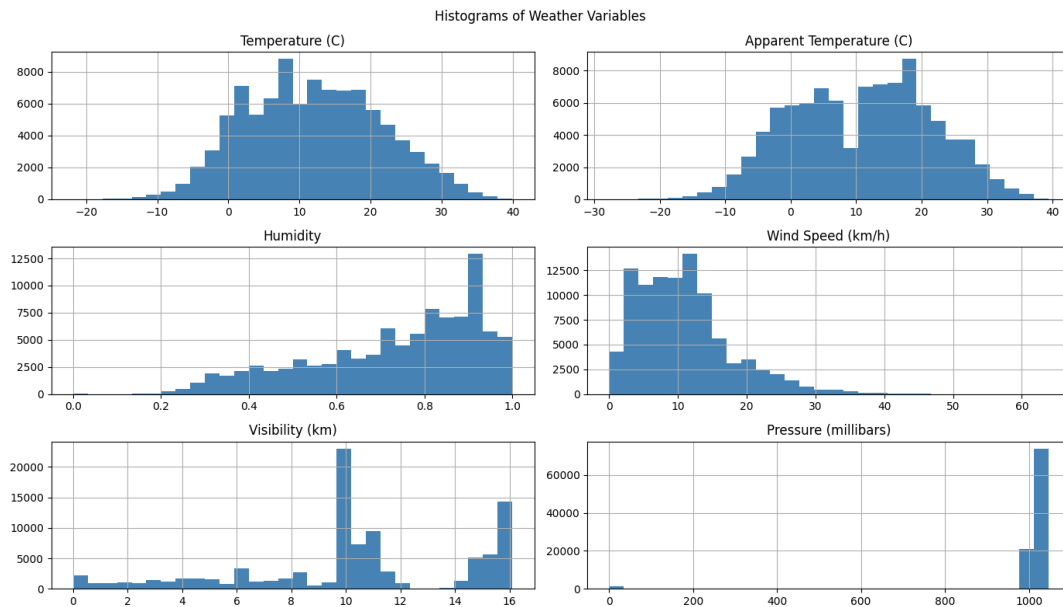
Variable	Skewness	Direction	Interpretation
Temperature(C)	11.93	Symmetric	Normally distributed
Apparent Temperature(C)	10.86	Symmetric	Very similar distribution to actual temperature.
Humidity	0.73	Left skewed	Most readings are high
Wind Speed(km/h)	10.81	Right skewed	Most days are calm
Visibility(km)	10.35	Left skewed	Most readings show good visibility
Pressure(mb)	1003.24	Ext.High	Zero readings (might be errors)

Key observation from the above stats:

- Wind Speed shows the greatest number of extreme outliers which represents the storm events.
- Pressure shows 1 extreme low outlier near zero.

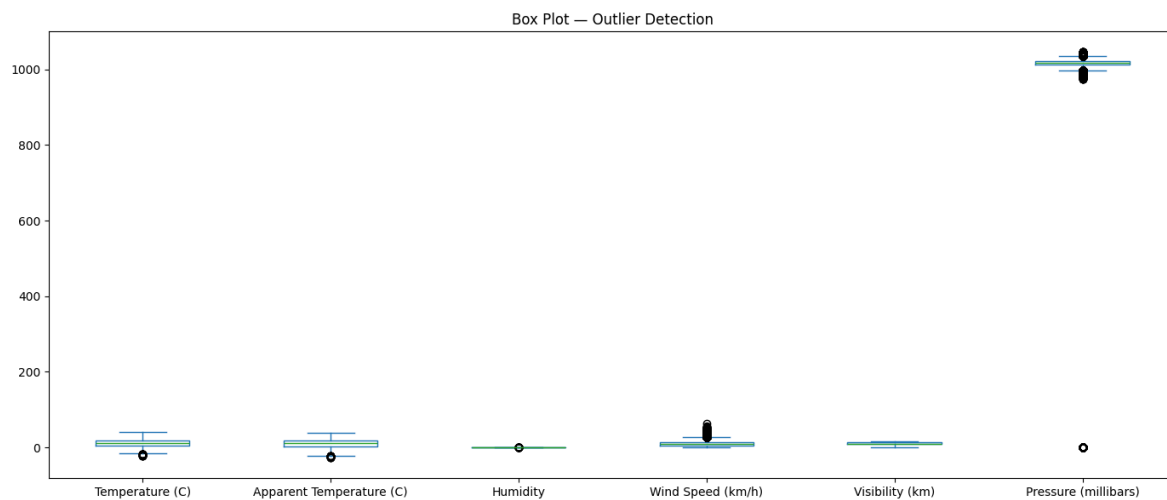
SECTION 4: Chart analysis

1) HISTOGRAM - Distribution of variables:-



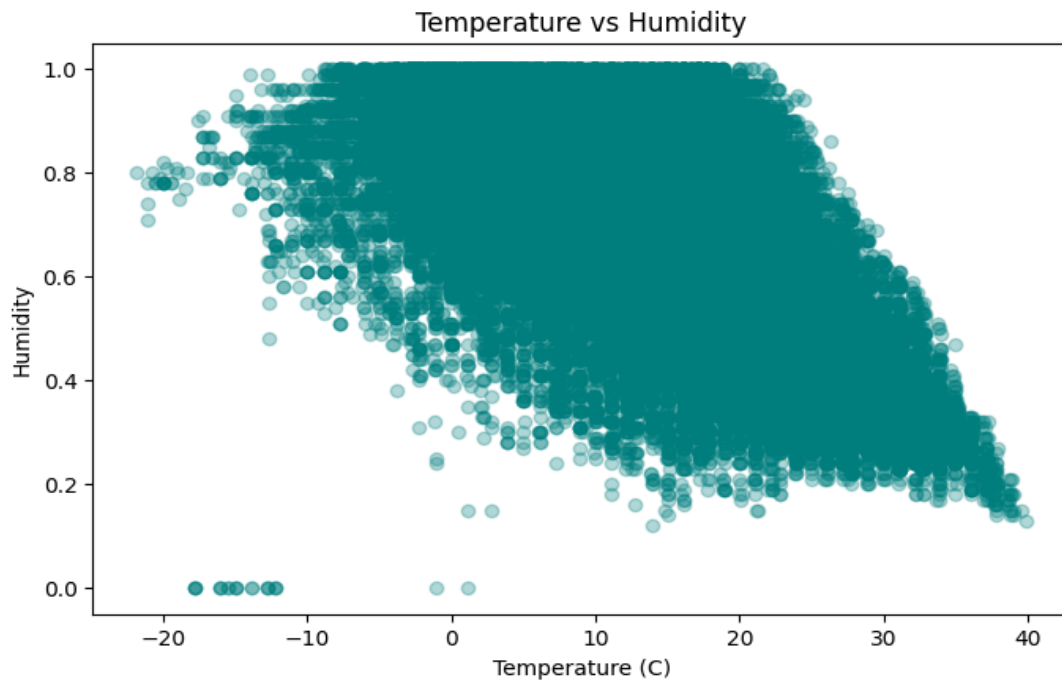
- Temperature is evenly spread (most days fall between 5°C & 20°C).
- Humidity is mostly high (majority of the readings were above 0.6).
- Wind Speed is mostly low (calm days are far more than common storm days).

2) Box Plot : Outlier detection:-



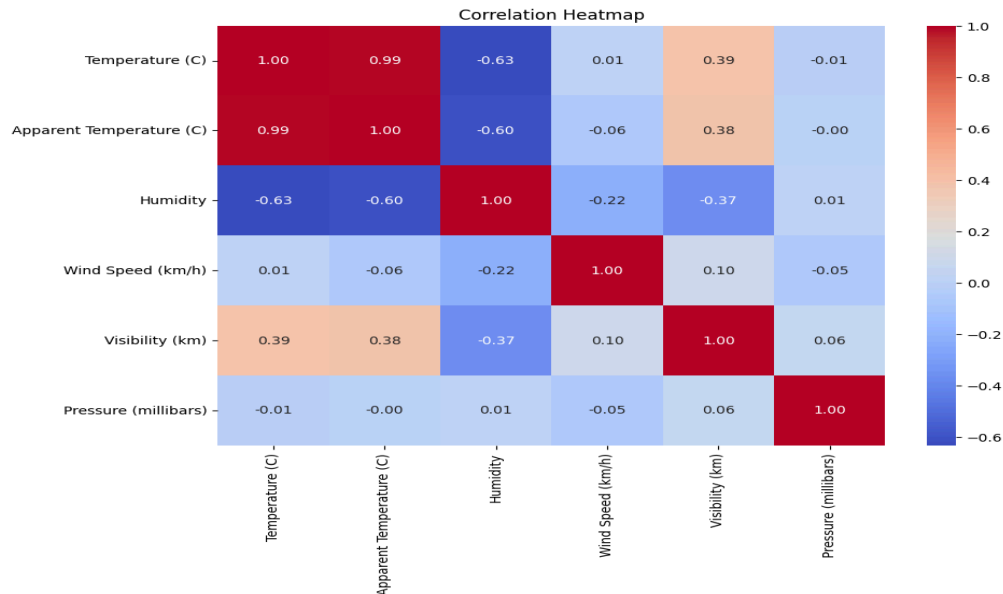
- Wind Speed has the most outliers (some days were recorded speeds above 40 km/h).
- Pressure has one extreme low outlier near zero (likely a sensor error).
- Temperature and Humidity have very few outliers (data is mostly consistent).

3) Scatter Plot - Temperature vs Humidity:-



- Clear downward pattern (as temperature goes up the humidity goes down).
- Cold days below 0°C tend to be very humid.
- Hot days above 30°C tend to be much dry.

4) Correlation Heatmap:-



- Temperature and Apparent Temperature: 0.99 (almost identical, expected).
- Temperature and Humidity: -0.63 (strong negative link, hotter means drier).
- Temperature and Visibility: 0.39 (warmer days tend to be clearer).
- Pressure: near zero with everything (behaves independently).

Key findings:-

According to my knowledge I found the following areas ;



Conclusion:

Hence from the above analysis of all the learnings and reports of 96,453 , we have successfully identified the key stats properties , distribution patterns across different sections , outliers and variables. Thus the strongest relationship found is the - correlation between temp & humidity (-0.63). Temperature was normally distributed with expected seasonal variation while the wind speed outliers represents the real storms where the pressure behaves as an independent variable with no significance from the use case and learnings so far from the calculations.